METRIC

DOD-P-24631/1A(SH)
16 October 1987
SUPERSEDING
DOD-P-24631/1(SH)
13 April 1984

MILITARY SPECIFICATION SHEET

PAINT, EPOXY, BLACK, NAVY FORMULA 184, TYPE I (METRIC)

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the paint described herein shall consist of this specification and the latest issue of DOD-P-24631(SH).

Formula: This formula covers black epoxy paint designated Navy formula 184 for exterior use. The paint shall consist of the ingredients specified in the quantities specified.

AMSC N/A FSC 8010

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TABLE I. Navy formula 184 - component A.

	Component A	Pounds	Kilograms	Gallons[1/]	Liters[1/]
Curing agent	2216A (clear amber), 3M Company	1496.0	678.5	182.93	692.30
Carbon black	Raven 1255 beads, Columbian Carbon	28.0	12.7	1.86	6.96
Titanium dioxide	ASTM D 476, type III or IV	8.8	4.0	0.26	1.00
Magnesium silicate	MIL-P-15173, type A	945.6	428.9	41.97	158.80
Aromatic solvent	HI SOL 10, Ashland	144.0	65.3	19.72	74.58
n-Butyl alcohol	ASTM D 304	1197.6	543.2	176.96	669.81
Ethylene glycol mono ethyl ether	TT-E-781	616.0	279.4	79.28	300.05
Ethylene glycol mono butyl ether	TT-E-776	103.2	46.8	13.71	51.92
Thickener	Bentone 38, NL Ind.	28.8	13.1	2.03	7.73
Thickener activator	Propylene carbonate, Jefferson Chemical	9.6	4.4	0.97	3.68
Dispersant	Tenlo 70, Diamond Shamrock	2.4	1.1	.29	1.10
		4580.0	2077.4	519.99	1967.96

[1/] Provided for guidance only - actual values are dependent on variation in raw materials.

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TABLE II. Navy formula 184 - component B.

	Component B	Pounds	Kilograms	Gallons[1/]	Liters[1/]
Epoxy resin	Scotch Weld 2216B Clear, 3M Company	1362.97	618.23	142.02	537.45
Carbon black	Raven 1255 Beads, Columbian Carbon	24.06	10.93	1.60	6.10
Titanium dioxide	ASTM D 476, type III or IV	7.99	3.57	0.24	0.89
Magnesium silicate	MIL-P-15173, type A	865.91	392.81	38.43	145.44
Aromatic solvent	HI SOL 10, Ashland	174.76	79.24	23.94	90.55
n-Butyl acetate	TT-B-838	1448.83	657.74	196.62	744.84
Ethylene glycol mono ethyl ether	TT-E-781	745.70	338.27	95.98	363.24
Ethylene glycol mono butyl ether	TT-E-776	125.05	56.75	16.61	62.99
Thickener	Bentone 38, NL Ind.	40.88	18.50	2.88	10.83
Thickener activator	Propylene Carbonate, Jefferson Chemical	13.66	6.20	1.38	5.17
Dispersant	Tenlo 70, Diamond Shamrock	2.42	1.05	0.29	1.05
		4812.24	2183.28	519.99	1968.56

[1/] Provided for guidance only - actual values are dependent on variation in raw materials.

NOTES:

1. The curing agent shall be Scotch Weld 2216A (transparent) available from 3M Company, 3M Center, 2501 Hudson Road, St. Paul, MN, or equivalent (see figure 1 of this specification sheet).
2. The carbon black shall be "Raven 1255 beads" available from Columbian Carbon, Akron, OH, or equivalent.
3. The solvent shall be any 97-99 percent aromatic with a flash point of 100-110 degrees Fahrenheit (°F) (38-43 degrees Celsius (°C)).
4. The thickener shall be "Bentone 38" available from N.L. Industries, Inc., Hightstown, NJ, or equivalent.
5. The thickener activator shall be "propylene carbonate" available from Jefferson Chemical, Shaker Heights, OH, or equivalent.
6. The dispersant shall be "Tenlo 70" available from Diamond Shamrock, Morristown, NJ, or equivalent.
7. The epoxy resin shall be Scotch Weld 2216B (clear amber) available from 3M Company, 3M Center, 2501 Hudson Road, St. Paul, MN, or equivalent (see figure 2 of this specification sheet).
8. The magnesium silicate shall be used in accordance with MIL-P-15173, type A and shall be asbestos-free.

Use of alternative ingredients in this formula shall have prior approval by the Naval Sea Systems Command. Approval will be based on review of data showing equivalent physical and chemical characteristics to the specified ingredient. It will be necessary to demonstrate that the paint compound made using the alternative ingredient will conform to all requirements of this specification.

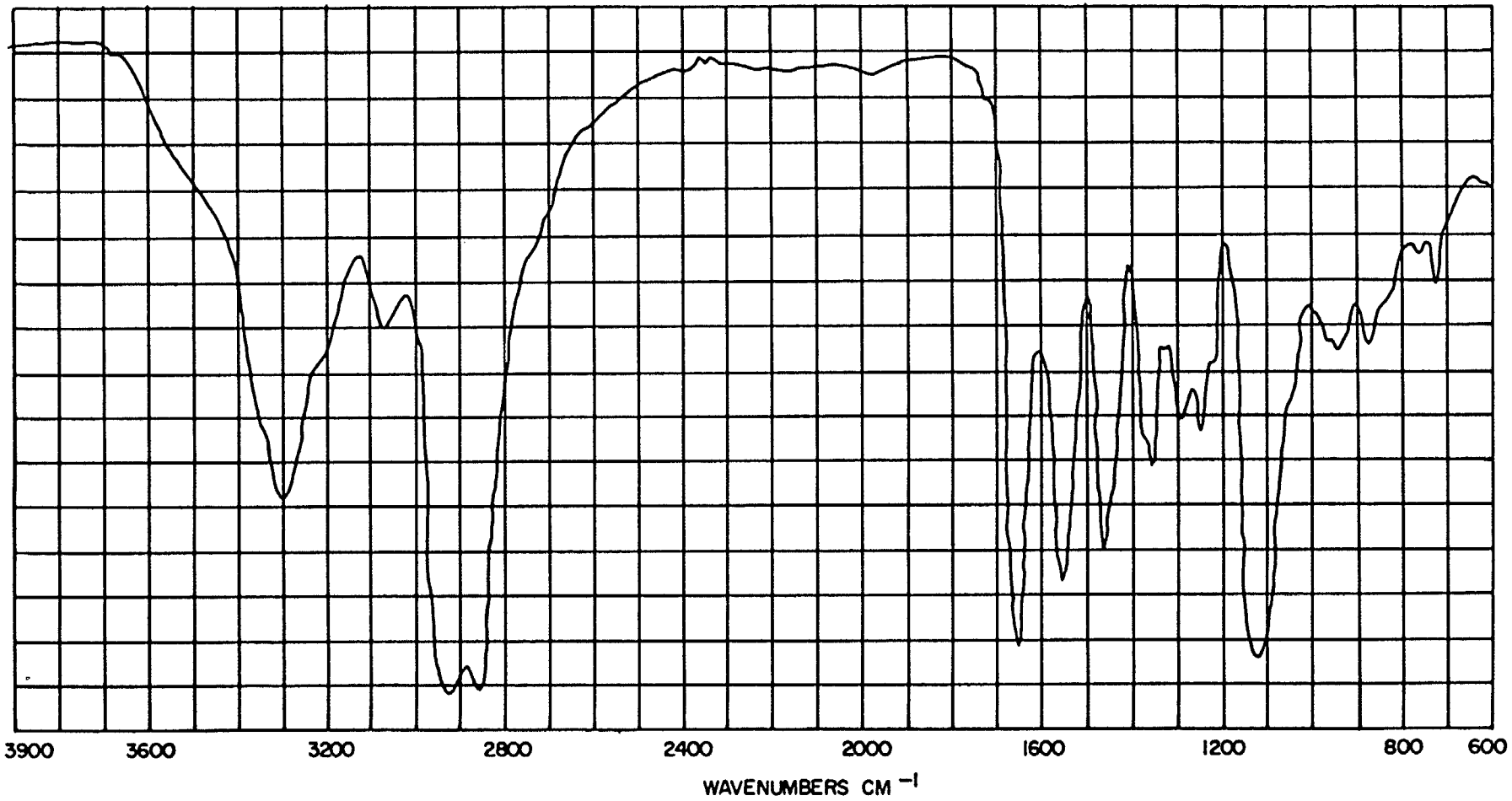
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TABLE III. Quantitative requirements.

Requirements	Component A		Component B		Mixed component	
	Min	Max	Min	Max	Min	Max
Pigment:						
Percent by mass of paint	19.45	23.45	16.66	20.66		
Percent by volume of paint	6.48	10.48	5.74	9.74		
Volatiles:						
Percent by mass of paint	43.21	47.21	62.33	66.33		
Percent by volume of paint	53.90	57.90	36.62	40.62		
Nonvolatiles:						
Percent by mass of paint	31.34	35.34	27.22	31.22		
Percent by volume of paint	33.63	37.63	25.92	29.62		
Consistency, Krebs units					60.0	----
Mass per unit volume:						
lbs/gallon	8.56	9.06	9.00	9.50	8.78	9.28
Kg/L	1.03	1.09	1.08	1.14	1.05	1.11
Fineness of grind (Hegman)					6.0	----
Flash point, Pensky-Martens						
°C					27.0	----
°F					80.0	----
Wet reflectance color (Rw), percent					0.8	1.02
Set-to-touch time:						
hours at 23°C (73°F)					6.0	8.0
Dry-hard hours at 23°C (73°F)					----	24.0
Titanium dioxide, percent of pigment	0.18		0.18			
Pot life, hours at 23°C (73°F)					6.0	----
Gloss, 16°C (60°F) specular, percent					----	10.0
Contrast ratio					0.9	

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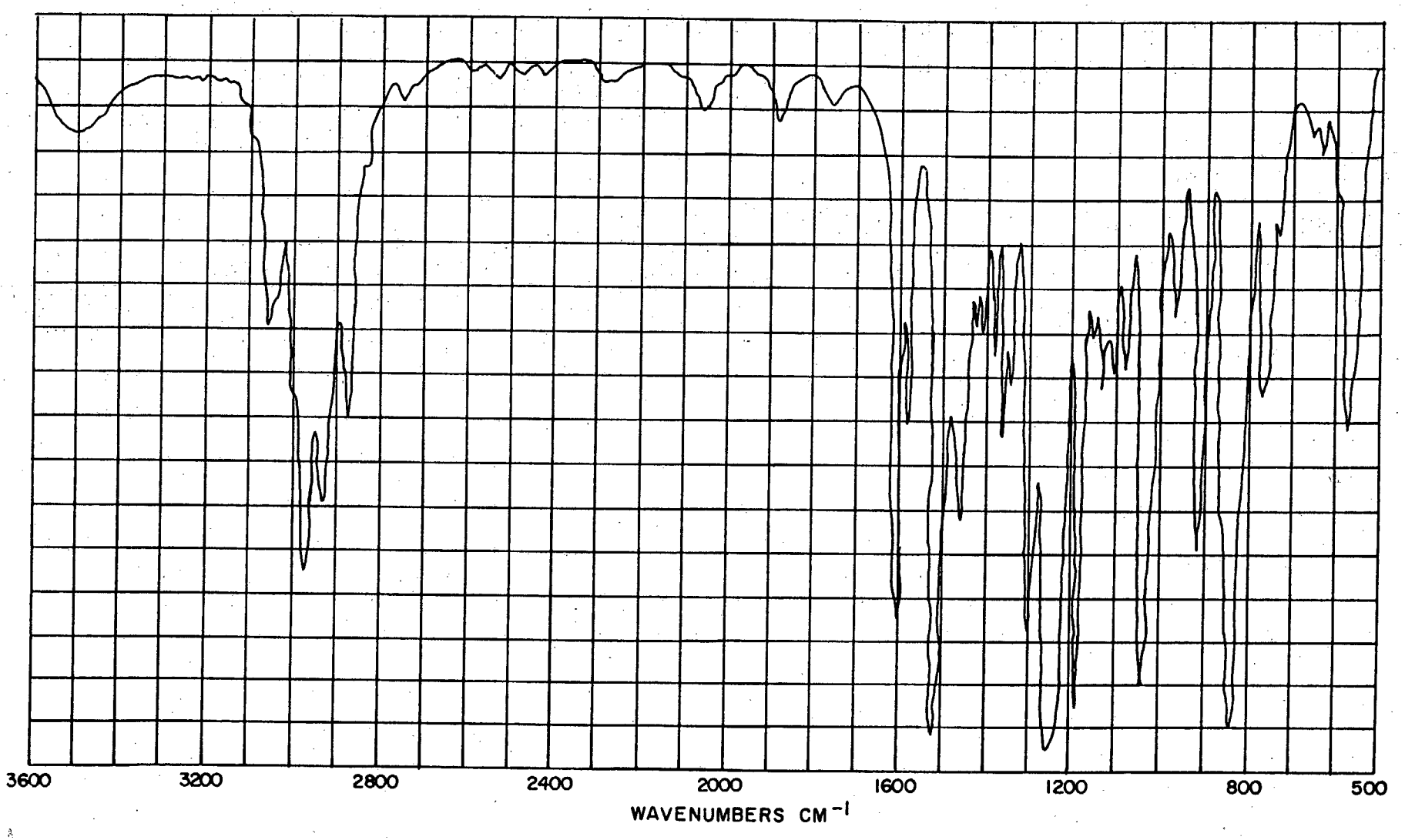
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SH 12139

FIGURE 1. Infrared spectrogram of Scotch Weld 2216A (aliphatic amine).

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SH 12140

FIGURE 2. Infrared spectrogram of Scotch Weld 2216B (epoxy resin).

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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